Technical review

Item #1 · 01-clean-rsa.eml · Run ID: 20261113T232800Z-08b8d1

Path	`R:\evidence\mail\01-clean-rsa.eml`
Size (bytes)	1464
Last modified (UTC)	2026-11-12T06:00:00Z
Extension	.eml
Detected type (signature)	Email message (RFC 5322)
Status	No issues found
Case	RV-SAMPLE-001
Evidence ID	E0015
Case registry	Verified: identical to the registered evidence
Registered at intake (UTC)	2026-11-13T23:27:52.795Z

Hashes

sha256	`d774e6fb20d569075f2767ee4dd92136071f6c2fa81ce0979600069b89596f59`
md5	`3b910e8841d5a4b560a2157d9595b08c`

Findings

[Informational] DKIM signature verifies (DKIM_SIGNATURE_VALID)

What it shows	The DKIM signature of example.org (selector s2026, rsa-sha256) verifies with the key from key_file: the signed header fields and the body are unchanged since signing.
What it does not prove	It shows only that whoever held the private key of example.org signed this content; it does not show who wrote it, that the From address is truthful, or that unsigned header fields are original.

[Informational] Receiving server's checks recorded (AUTH_RESULTS_RECORDED)

What it shows	1 Authentication-Results header(s); the top one, written by mx.acme.example, states: dkim=pass, spf=pass.
What it does not prove	These are claims written into the message by a server; anyone handling the file could have added or edited them, and they were not re-checked.

Email analysis

Message-ID	20261110090000.1@example.org
From	Bruno Proveedor <bruno@example.org>
Return-Path	bruno@example.org
Subject	Acta de la reunión de noviembre
Date header	Tue, 10 Nov 2026 09:00:00 +0100 (2026-11-10T08:00:00Z)
DKIM mode / key file	offline · R:\inputs\dkim-keys.txt · 4 key(s)

Received hops (first server = 1)

Hop 1	From host: [192.0.2.10] · Received by: smtp.example.org · Date stamped: 2026-11-10T08:00:00Z

Hop 2	From host: smtp.example.org · Received by: mx.acme.example · Date stamped: 2026-11-10T08:00:04Z

DKIM signatures (offline verification)

#1 d=example.org s=s2026	Result: pass · Algorithm: rsa-sha256 · Key source: key_file (line 3)
ARC chain	none (0)

Authentication-Results (as stated by the servers)

mx.acme.example	dkim=pass, spf=pass

Files written by the analysis

headers.json (headers)	md5 b8d915a5a101147d18643b4100472c7f · sha256 dc501dee75e7305e1f879b1e21decbfd3bd0904d5274545b3fd6a06f7951b850
dkim.json (dkim)	md5 64883f418b95f3dae626ab496ea1c713 · sha256 a69761958a4c9227a1a2ccedaafc73e4a5d92c7045688a8daaaf309db47a0ae6

Method

Each file was read once to compute its digests, then analysed as email: .eml as an RFC 5322 message, .msg through its MAPI properties, and each message of an mbox as an item of its own (each in a separate worker process with time and memory limits when worker isolation is on, as recorded in the audit log). The header section, Received hops and dates were read and compared; DKIM and ARC signatures were verified offline with the public keys the examiner supplied (no DNS query unless online mode was chosen); Authentication-Results were recorded as stated evidence; attachments were hashed; and BEC heuristics compared the addresses with the organisation's own domains. Every step and every file written is recorded in the hash-chained audit log.

Audit trail

Audit entries	#6 – #12
Log folder	`R:\case\runs\rv-mailauth\20261113T232800Z-08b8d1\logs`
Digital signature	This PDF is not digitally signed.

Limitations

- DKIM results depend on the public keys supplied: a check is only meaningful with the key that was valid when the message was sent. Keys are rotated, so a current key can fail a genuine message and a missing key leaves the signature unchecked.
- A missing DKIM signature is common and proves nothing; a valid signature covers only the signed header fields and the body, not who wrote the message.
- Outlook .msg files do not keep the original message: the body is re-encoded (so DKIM cannot be checked) and the original header section is kept only when transport headers were stored.
- Lookalike-domain, Reply-To, display-name and payment-wording checks are heuristics with false positives and false negatives; the organisational domain comes from the bundled Public Suffix List, whose version is recorded in environment.json and may differ from the list in force when the message was sent.
- Received, Date and Authentication-Results headers are text that servers and clients wrote; unsigned headers can be added or edited by anyone who handled the file.
- In an mbox, body lines starting with "From " are escaped with '>'; un-escaping follows the selected variant (mboxrd by default) and cannot always restore the original line, which can affect DKIM body hashes.
- A result without findings does not prove that a file is authentic or unaltered before acquisition.
- Modification times come from the file system of the examined copy and may differ from the original source.
- Type detection uses a limited table of common file signatures; unknown types are reported as unknown, not as

suspicious.

- A trusted timestamp proves the audit log existed in this exact form at the stated time; it does not prove that the examined copy matches the original source.